When Does RLVR Teach Physical Strategy? Measuring Simulator-Only Post-Training for CNC Planning

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Abstract

We present a single-run case study of reinforcement learning from verifiable rewards (RLVR) for CNC process planning, separating program executability, strict machining success, and plan structure. Qwen3-8B writes complete multi-setup G-code; an interpreter and voxel process model supply the only post-SFT training feedback. The verifier is audited with 166 checks and a separate 2.5D stock checker. Two exams probe strategies described in the prompt but absent from teacher trajectories: multi-tool roughing/finishing of one feature and re-grouping nonadjacent same-face features. GRPO improves modeled path validity ($86.0 \to 93.0\%$ IID; $59.0 \to 76.0\%$ on the setup exam) and mean verifier reward ($-0.417 \to -0.344$ validation). Strict success changes by only 0.5 percentage points on validation and IID; both exams remain at $0/100$ with zero positive strategy proxies. Individual repairs coexist with regressions. These results describe one seed and a short, non-thinking training regime; they do not distinguish limited exploration, SFT-induced behavior, generation length, and reward or credit-assignment effects. The contribution is an audited benchmark and a measured separation of outcomes, rather than general conditions for strategy acquisition. We release the frozen benchmark, verifier, and archived outputs for offline regrading.

1 Introduction

Foundation and world models increasingly support decision-making [Ha and Schmidhuber, 2018, Hafner et al., 2023, Yang et al., 2024, Bruce et al., 2024, Hu et al., 2023], but predictive fidelity alone does not establish reliable action. An agent can still gouge a part or collide on a rapid. It can also waste setups. We operationalize one aspect of physical understanding as choosing actions that respect stock geometry and tool-machine constraints. We ask when post-training against a physical grader adds process strategy beyond the demonstrations.

RLVR has improved language models where a checker can replace preference labels [Lambert et al., 2024, DeepSeek-AI, 2025, Shao et al., 2024]. The same recipe is now applied to physical constraints [Ji et al., 2025]. A reward can raise pass@1 while sharpening support the base policy already had [Yue et al., 2025, Yuan et al., 2026], so a gain on the grader is not, by itself, new strategy.

We measure the split on CNC process planning, where every cut irreversibly changes the stock. The input is a semantic feature list on fixed AL6061 stock, not STEP or drawings. The policy emits one multi-setup G-code document. An interpreter and voxel stock model grade interpreter acceptance, modeled path validity, and material state (Figure 2). Collision and tool reach are checked explicitly. We audit the grade with planted resolution films. A separate 2.5D engine and a 166-check battery provide further checks.

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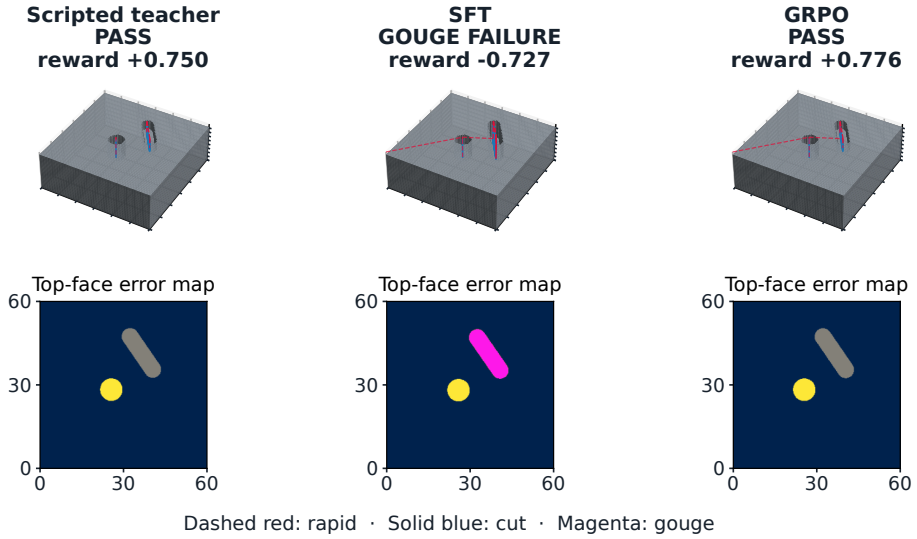


Figure 1: A rare local repair, not the mean effect. On validation seed 5177, the top row shows commanded toolpaths (rapids dashed); the bottom row projects removed thickness (dark blue: zero; gray: intermediate; yellow: greater removal), with red for required material remaining and magenta for protected material removed. Maps use 0.2 mm voxels; policy verdicts and rewards use the audit-plus-spot protocol. SFT gouges; GRPO repairs this part. Aggregate success moves only $18.0\% \rightarrow 18.5\%$ (Table 1).

The teacher is deliberately local. It uses one tool per feature and preserves feature-list setup order. It therefore never demonstrates two strategies the grader can reward. One is multi-tool roughing and finishing. The other reorders nonadjacent same-face features into one setup. Both rules are stated in the system prompt, but the teacher does not implement them. A $K{=}16$ parameter search can reduce overload on a small multi-tool subset, but it cannot produce either demonstration-absent composition (Appendix A). Closest work on G-code and CAD uses executable checks or formal proofs [Abdelaal et al., 2025, Lee, 2026, Mallis et al., 2026]. We report greedy SFT and simulator-only GRPO [Shao et al., 2024, Liu et al., 2025]. Two four-sample SFT controls provide context. They are any-of-four $\mathrm{pass@4}$ and reward-selected search-4.

Any-of-four $\mathrm{pass@4}$ measures whether four SFT samples expose a successful output; search-4 tests whether the coarse verifier selects one. Neither establishes that an unobserved behavior has zero probability under SFT. On both exams, both controls remain at $0/100$. GRPO improves path validity and mean reward, while exam success and strategy proxies remain zero. Occasional in-distribution repairs occur (Figure 1), but they are not representative. Code, data, and archived outputs are available at `https://github.com/pqori/CNC-RLVR`.

2 Task and verifier design

Task. Stock is $60 \times 60 \times 20$ mm AL6061. Milled features and drilled holes may occur on any face. The input is face-local feature geometry. The policy emits a complete document without intermediate simulator feedback, using a documented G-code subset (Appendix B). It uses absolute metric coordinates and inline setup declarations. Core commands control tools and motion. The `M3` command starts the spindle. Spindle speed and feed are fixed by the selected tool and move geometry, not emitted by the policy. Words outside the dialect and unsupported modal state fail with a line number.

Grade. Every move is swept through a 0.2 mm voxel stock model. Path validity requires accepted syntax and modeled motion with no rapid or holder collision. Removing protected material outside the tolerance envelope is a gouge. Tool reach and modeled load receive continuous severity

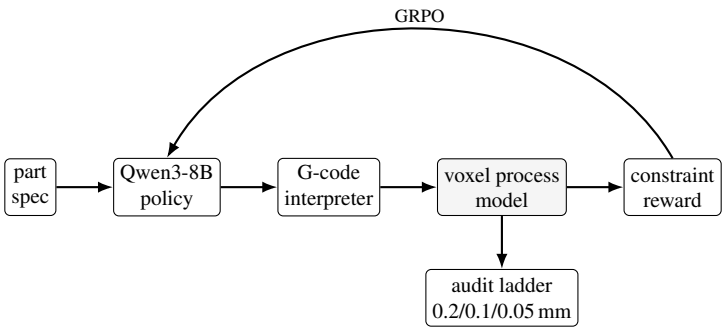


Figure 2: Product spec to G-code document to interpreter and voxel process model. GRPO uses the 0.2 mm grade with triggered 0.05 mm frontier checks; evaluation independently applies the fixed 0.1 mm audit and rechecks every audit-clean output at 0.05 mm.

scores. Ramp angle does too. Any violation fails the corresponding safety gate, although simulation continues to provide partial credit. We report

$$\texttt{valid} = \text{all safety gates pass}, \qquad \texttt{success} = \texttt{valid} \wedge c \geq 0.98 \wedge L_\infty \leq 0.1 \text{ mm},$$

where c is required-volume clearance and L_∞ is worst residual height. Evaluation repeats the sweep at 0.1 mm. Every audit-clean output is then rechecked at 0.05 mm. Appendix C gives the full objective and severity terms. Appendix F reports 98 exact cross-engine matches and a $166/166$ audit battery.

Designed demonstration gaps. The freeze has 1,999 teacher examples. Validation and IID evaluation each contain 200 parts. Each exam contains 100. The teacher uses one tool per feature and feature-list setup order. The multi-tool exam rewards using a large mill to rough and a slender mill to finish tight corners; the teacher's single-tool schedule overloads on every case. The setup exam interleaves three orientations; grouping by face reduces five setups to three. Local parameter search can reduce the multi-tool teacher's overload, but cannot compose roughing and finishing tools. On the setup exam it recovers only 0.003 of the reference's 0.202 reward gain (Appendix A). Modeled load is not exact geometry, so the multi-tool axis is an auxiliary composition instrument; depth and tool selection also change.

3 Training and evaluation

We fine-tune Qwen3-8B [Yang et al., 2025] for two epochs on the teacher corpus with LoRA [Hu et al., 2022]. That SFT policy is the imitation baseline and the GRPO start. GRPO samples four documents per prompt and uses group-relative advantages without further demonstrations. Each update uses eight prompts with four samples per prompt. Fresh GRPO tasks come from the training generator, with neither exam family explicitly mixed in. The run uses one seed with thinking disabled. GRPO caps outputs at $8{,}192$ tokens; the released evaluation configuration permits $24{,}576$ (Appendix E). Coarse-clean, faster-than-reference rollouts are regraded at 0.05 mm. From four samples of the same SFT checkpoint, we report both any-of-four $\mathrm{pass@4}$ and search-4, which selects the highest 0.2 mm reward. These are test-time controls (Appendix E). Headline numbers are greedy. Checkpoint selection is on validation (step 50 of 100).

4 Results

SFT baseline. SFT is path-valid on 88.5% of validation and 86.0% of IID parts. Strict success is only $36/200$ and $30/200$, respectively (Appendix A). Both exams remain at $0/100$, with neither strategy proxy observed.

GRPO changes path validity and reward more than strict success. Table 1 shows higher path validity on every split and higher in-distribution reward. Paired reward also improves when both

Table 1: SFT greedy versus GRPO greedy at the validation-selected checkpoint (step 50). Path-valid excludes hard path failures. Reward and success use the audit-plus-spot protocol. Two-mill is a setup-level proxy, not feature attribution.

split	metric	SFT greedy	GRPO greedy
validation (n=200)	path-valid%	88.5	91.5
	reward	-0.417	-0.344
	success%	18.0	18.5
	gouge / n	105	112
	two-mill%	7.0	7.0
IID test (n=200)	path-valid%	86.0	93.0
	reward	-0.403	-0.339
	success%	15.0	15.5
	gouge / n	92	101
	two-mill%	3.5	9.0
OOT multi-tool (n=100)	path-valid%	89.0	96.0
	reward	-0.908	-0.906
	success%	0.0	0.0
	two-mill%	0.0	0.0
OOT setups (n=100)	path-valid%	59.0	76.0
	reward	-0.906	-0.884
	success%	0.0	0.0
	G-code fails	21	10
	fewer-setups%	0.0	0.0

outputs are already path-valid. The changes are $+0.066$ on validation and $+0.044$ on IID, so transitions into path validity do not explain the full gain. Strict success changes by one part on each in-distribution split. Repairs are nearly balanced by regressions (Appendix A). Raw gouge counts also rise; unconditional counts do not establish a higher conditional gouging rate. Neither greedy GRPO nor either four-sample control achieves exam success or a positive exam proxy. Figure 1 is therefore a local repair, not the aggregate result.

5 Discussion

The result separates three outcomes. GRPO produces more path-valid programs and raises verifier reward, including among pairs already path-valid under both policies. Strict part success changes little, and raw gouge counts rise. A higher scalar grade therefore does not by itself establish acquisition of a planning strategy.

Both probes require coordinated decisions across a long document. Multi-tool machining links tool choices within a feature; setup grouping requires nonlocal ordering. Terminal reward gives no direct attribution to either decision. Although the verifier accepts scripted references, the observed zeros cannot distinguish limited sampling support, SFT suppression, the generation cap, reward scaling, or credit assignment. The strategies are prompted, so these are tests of elicitation and composition rather than discovery of an unstated rule.

The evidence is conditional on one validation-selected checkpoint from one training seed. Reported differences are descriptive, and neither zero exam success nor four unsuccessful samples establishes absent policy support. The two-mill proxy can count tools serving different features, while the setup proxy is gated on feasibility. Base-model, larger-sample, reasoning-enabled, length-controlled, and additional-seed comparisons remain unrun (Appendix E). Evaluation should therefore report physical outcomes and inspect plan structure alongside reward. The model's physical scope is stated in Appendix G.

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Table 2: Scripted baselines (n=20 per exam axis; parameter search uses a five-task subset). Search can reduce overload without composing the demonstration-absent strategy. These are not trained-policy results.

axis	plan	outcome	reward
multi-tool	teacher	0/20	0.308
multi-tool	geometry search (K=16)	5/5 feasible	0.515
multi-tool	two-tool reference	20/20	0.750
setups	teacher (5 setups)	20/20	0.650
setups	parameter search (K=16)	5/5	0.653
setups	grouped reference (3 setups)	20/20	0.852

Table 3: Greedy SFT on the frozen splits. "Final" applies the audit-plus-spot protocol. `valid` means all safety gates pass. Path-valid rates are higher (88.5% / 86.0% IID-family).

split	`valid`%	success@0.2	success final	reward
validation	35.5	20.0	18.0	−0.417
IID test	39.5	18.0	15.0	−0.403
OOT multi-tool	0.0	0.0	0.0	−0.908
OOT setups	1.0	0.0	0.0	−0.906

A Additional tables and figures

Paired diagnostics. Validation contains two SFT-failure to GRPO-success repairs and one regression. IID contains three repairs and two regressions. Among pairs that are path-valid under both policies, mean reward changes by $+0.066$ on validation and $+0.044$ on IID. The two-mill proxy requires two distinct mills to remove stock in one setup. They may serve different features, so the IID increase from 3.5% to 9.0% does not establish same-feature roughing/finishing. The fewer-setups proxy requires a feasible program with a lower setup count than its teacher; unsuccessful regrouping attempts can therefore be missed. Neither proxy alone establishes strategy completion.

B Full task and G-code dialect

Coordinate frames. Features use face-local (u, v) coordinates in the frame seen by the tool after reorientation. A setup declaration selects one of six axis-aligned faces. The X, Y coordinates are face-local. The entered stock face defines $Z = 0$, with negative Z inside the material. One interpreter therefore handles both side and top machining.

Dialect. A document begins with `G21 G90`. The `(SETUP F)` directive opens a setup, while `T<n> M6` changes tools. The spindle starts with `M3`. Linear motion uses `G0` or `G1`. `G2` and `G3` express I/J-offset arcs. Equal endpoints denote a full circle, while a Z word creates a helix. Coordinates are modal. RPM and feed are injected. The document ends with `M30`. Unsupported controller features are rejected. These include cutter compensation and canned cycles. Incremental coordinates and R-form arcs are also rejected. Unknown words fail with a line number. Drills cut only axially. Endmills may ramp or helix but may not plunge.

```
G21 G90
(SETUP Z+)
T8 M6
M3
G0 X18.000 Y22.000 Z2.000
G3 X20.400 Y22.000 Z-1.050 I1.200 J0.000
G3 I-1.200 J0.000 Z-2.100
G1 X24.000
G0 Z2.000
M30
```

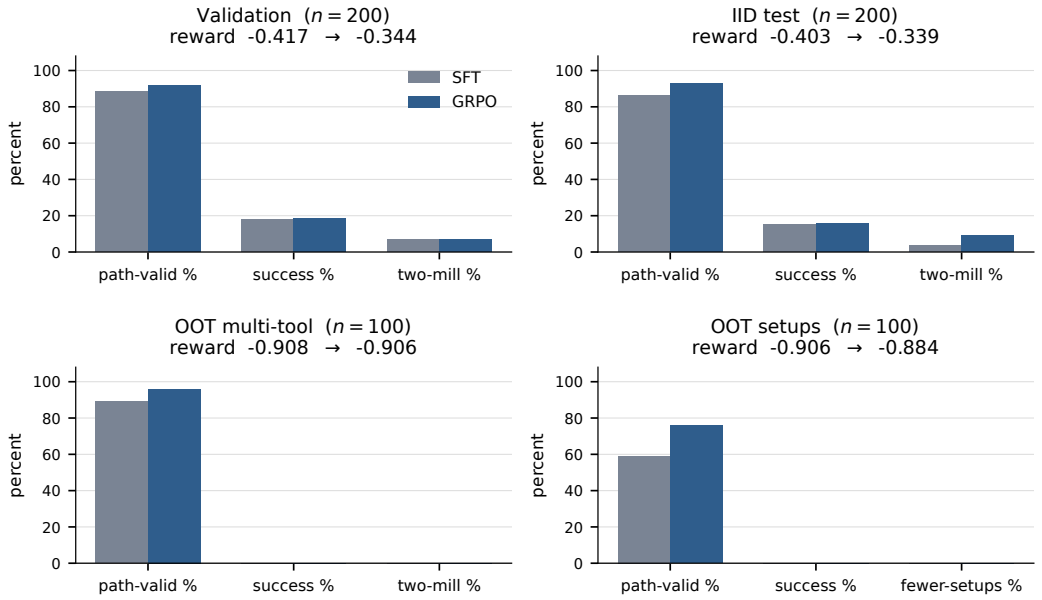


Figure 3: Where the process-model score moves. Path validity and reward improve. Success changes little; exam proxies remain zero. Two-mill means two mills cut in one setup; it is not the multi-tool exam name. Numbers are Table 1.

Fixed cutting schedule. Material is fixed to AL6061. Each tool has one canonical spindle speed/feed pair. Descending mill entries use half feed, while flat cuts use full feed.

tool	rpm	feed	tool	rpm	feed
EM3/EM3L	29,178	1,400.5	EM6	14,589	2,100.8
EM10	8,754	2,101.0	EM12	7,295	2,801.3
D3	9,549	458.4	D5/D8	5,730/3,581	458.4

Feeds are in mm/min. The interpreter injects these values from tool choice.

C Verifier implementation and objective

Voxel sweep. The training engine stores a $300 \times 300 \times 100$ byte grid at 0.2 mm resolution. Each normalized motion primitive is sampled with an error-bounded chord spacing; only a localized neighborhood of voxels is tested and subtracted. Required-removal and must-keep masks are generated independently from feature geometry. Side operations require a true 3D state. A cross-hole can create solid–air–solid columns that a single heightfield cannot represent.

Geometric and controller checks. Controller checks cover parsing and modal state. They also enforce coordinate bounds and setup limits. Geometric checks detect rapid and holder collisions. They enforce face access and tool compatibility. Entry geometry and completion are checked separately. Teacher documents must round-trip through emit–parse–emit byte-identically and produce identical motion primitives and metrics.

Load model. The current code measures newly removed volume over one-second cutting windows. It computes $\mathrm{MRR} = V/t$, where t is the union of newly removed voxel-center contact intervals. This excludes zero-removal recuts and air-feed padding. Equivalent G1 segmentation leaves the result unchanged. Cutting power is $P = k_c \mathrm{MRR}$. Mean force is $F = P/v_c$. End-mill capacity uses a cantilever-deflection envelope. Spindle power provides a second cap. With $u = \max(F/F_{\max}, P/P_{\max})$, evaluation reports $u > 1$ as load-unsafe and never counts it correct. Training continues the counterfactual path after overload and grades the severity continuously.

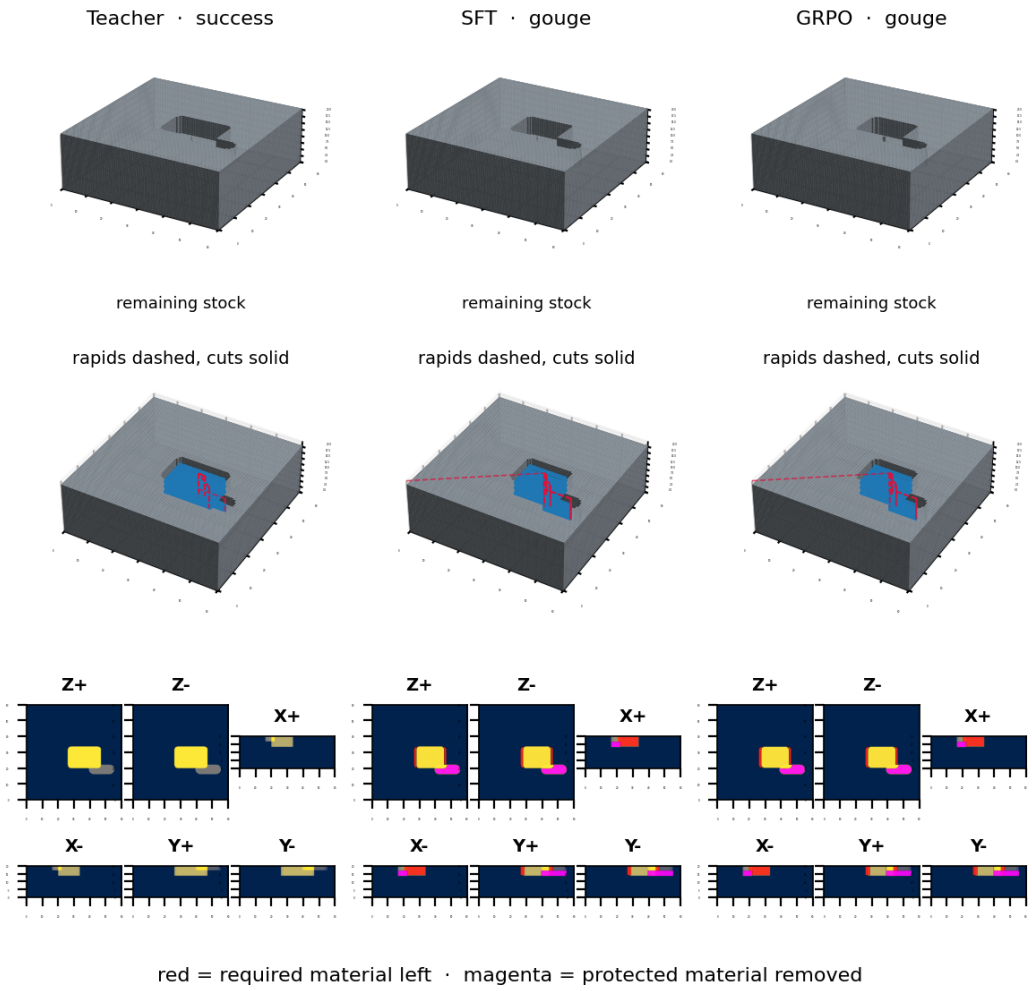


Figure 4: Typical in-distribution programs (IID seed 8050). SFT and GRPO are both path-valid and both gouge. Leftover is red; gouge is magenta.

Training reward. Reward v9 defines normalized ramp-slope utilization $u_r = \tan(\theta_{\max})/\tan(5°)$. Evaluation requires $u_r \leq 1$, but, as with load, simulation continues after an excess so training retains a severity signal. The reward first computes

$$\begin{aligned} q = c &- 0.25\,\mathrm{clip}_{[0,3]}(t/t_{\mathrm{ref}}) - 0.60\,\mathbf{1}[L_\infty > 0.1] \\ &- 2\max(0, 0.98 - c) - 0.05\min(s_{\mathrm{excess}}, 3) \\ &- 0.05\phi(u) - 0.05\phi(u_r) - 1.54\phi(u_{\mathrm{reach}}) - 0.06\phi(g), \end{aligned}$$

$$r = \begin{cases} -1, & \text{path-invalid hard failure}, \\ q, & q \geq -0.90, \\ -0.898769 + 0.001368q, & q < -0.90. \end{cases}$$

Here t_{ref} is the teacher cycle time when the teacher is feasible. The multi-tool reference supplies it otherwise. The term s_{excess} counts setups beyond the number of required feature faces. Path-invalid failures include malformed controller state and hard motion failures. Rapid or holder collision is a hard motion failure, as is illegal tool motion. For $z = \max(0, (u - 0.8)/0.2)$,

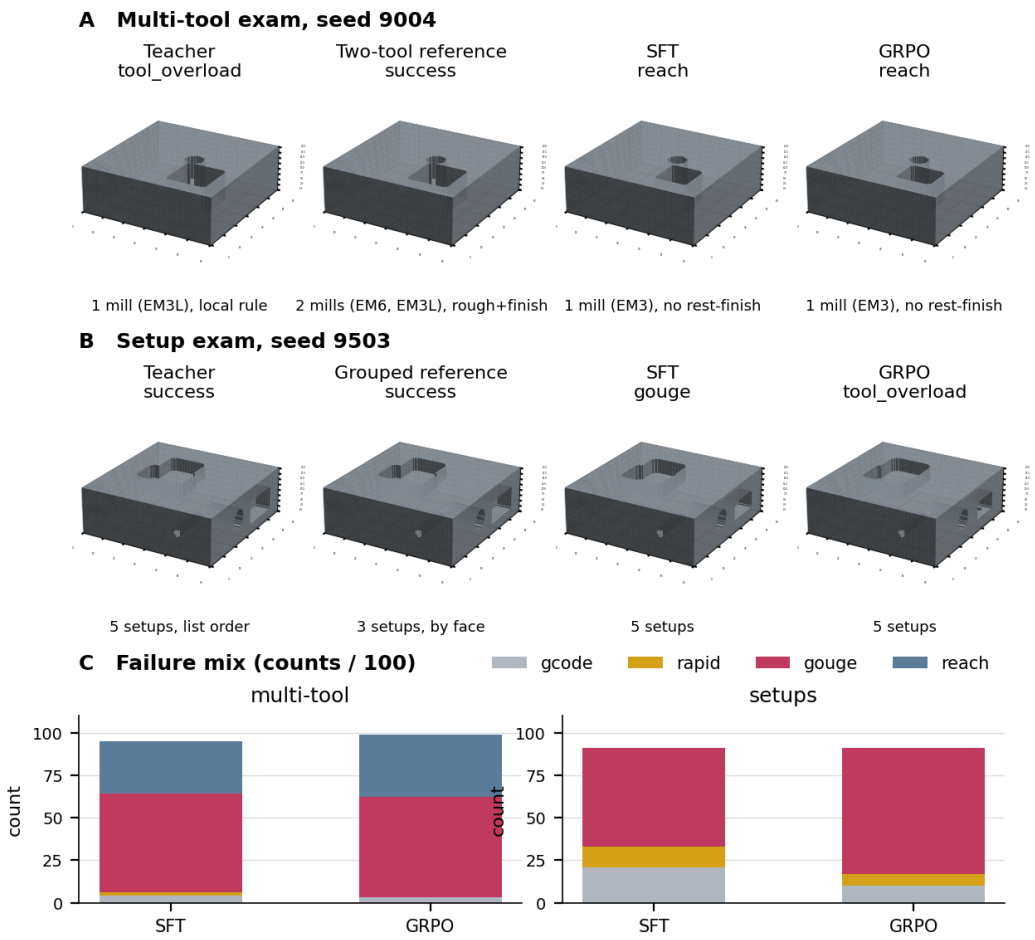


Figure 5: The exams ask for a different plan. Panel A shows seed 9004. The teacher overloads one long mill, while the two-tool reference succeeds. SFT and GRPO still use one mill. Panel B shows seed 9503. Grouping cuts five setups to three, while both policies keep five. Panel C shows the failure mix.

$\phi(u) = \min(20, z^2)$ for $z \leq 1$, and $\phi(u) = \min(20, 2z - 1)$ otherwise; $\phi(u_r)$ uses the identical dimensionless shape. Ramps through roughly 4° are unpenalized. The penalty is 0.05 at the 5° limit and remains continuous above it. Pure vertical non-center-cutting plunges remain hard failures. The load coefficient is the smallest 0.05-grid value above a 10% margin on paired intervention calibration. Safe candidates dominated unsafe counterparts on 42/42 calibration tasks and 16/16 held-out tasks. Half strength passed 95.2% and 93.8%, respectively. For flute utilization $u_{\text{reach}} = d_{\max}/L_{\text{flute}}$, $\phi(u_{\text{reach}}) = \min(20, \max(0, u_{\text{reach}} - 1)/0.25)$, so every use through the exact flute limit is penalty-free. The calibrated reach coefficient 1.54 achieved 270/270 calibration and 405/405 held-out safe-plan dominance. Half strength passed only 94.4% of calibration pairs. Gouge severity g is the maximum physical penetration beyond the analytic dimensional-tolerance envelope, with $\phi(g) = \min(20, g/(0.1\,\text{mm}))$. Coefficient 0.06 is the smallest 0.01-grid value above a 10% margin on 1,160 paired calibration shortcuts and achieved 576/576 dominance on disjoint held-out geometries. Half strength achieved 86.1% and 94.1%, respectively. Tool wear is measured but excluded. Evaluation preserves all four modeled safety gates and reports their severities separately. The path-valid raw objective is bounded below by -37.46, which maps to -0.95; the canonical reference remains 0.75. The strictly increasing map preserves every path-valid plan ordering and leaves all rewards at or above -0.90 unchanged while guaranteeing that any executable partial trajectory beats malformed output at -1.

D Benchmark construction and reference ladder

Tasks are generated deterministically from integer seeds and regenerated during evaluation to detect prompt or RNG drift. Verified teacher pairs are retained only if they pass the frozen checker. The release contains the following splits.

split	tasks	teacher success	purpose
train	1,999	1,999	SFT
validation	200	200	model selection
IID evaluation	200	200	interpolation
OOT multi-tool	100	0 under frozen load	composition
OOT setups	100	100	sequencing

All 1,999 teacher plans use one tool per feature. In 541 plans, the feature order makes the teacher revisit a face rather than regroup nonadjacent same-face features. The locked Qwen3 tokenizer measures a maximum teacher completion of 19,737 tokens across the frozen splits. SHA-256 hashes cover every JSONL file. The reference ladder compares the teacher, a $K{=}16$ stepdown/stepover search that cannot alter strategy. Hand-constructed references provide the structural comparison.

Teacher-failure taxonomy. Failures first separate controller execution from physical outcomes. Physical outcomes then distinguish geometry from process efficiency. A gain limited to controller execution is not evidence of physical strategy.

E Training details

SFT. Qwen3-8B uses LoRA rank 64 with $\alpha = 128$ on all linear layers. Training uses learning rate 10^{-4} for two epochs. Loss applies only to completions. Thinking mode is disabled and prompt formatting is byte-identical across training and evaluation. We retain the Qwen3 tokenizer and causal language-model output head, with no CNC-specific tokens, vocabulary resizing, or separate action head.

GRPO. For each prompt, $K{=}4$ documents receive

$$A_i = r_i - \frac{1}{K}\sum_{j=1}^{K} r_j,$$

without standard-deviation normalization. The update uses rank-64 LoRA with clipping $\epsilon = 0.2$. The KL coefficient is 0.04. The learning rate is 10^{-5}. Each step uses eight prompts, and the run lasts at most 100 steps. Document decoding is unconstrained. The $8{,}192$-token training cap is below the frozen $19{,}737$-token teacher maximum. Cap-truncated outputs are incomplete and receive the parser-failure reward of -1. The imposed cap can therefore confound length with model capability. Evaluation permits $24{,}576$ output tokens in the released configuration; the $8{,}192$ limit applies to GRPO training. The teacher maximum does not establish a minimum solution length or a truncation rate. Original generation token IDs, stop reasons, and complete training rollouts are not included in the release, so training-time truncation frequencies cannot be recovered from the archived evaluation text. The run fixes model/data RNG seeds and writes immutable metadata plus per-rollout parse/path/load fields and coarse-versus-spot rewards.

Four-sample controls. SFT sampling uses temperature 1.0 with four candidates per task. Any-of-four $\mathrm{pass@4}$ records whether at least one candidate passes the audit-plus-spot protocol. Search-4 instead selects the candidate with highest $0.2\,\mathrm{mm}$ reward. Exact ties retain the first candidate. Any-of-four reaches 18.0% on validation and 16.5% on IID. Search-4 reaches 16.5% and 15.0%, respectively. Both controls remain at 0% on the exams. The selected step-50 policy had seen about 1,600 rollouts. The full run generated about 3,200 and took 8,304 seconds on the eight-GPU node.

Interpretation and unrun controls. We did not run equal-total-rollout RAFT [Dong et al., 2023] or a pre-SFT base measurement. Larger-sample support analysis, thinking-enabled or plan-then-code generation, operation-level advantages, and a grammar-locked typed arm also remain unrun.

The small group size and short training budget can limit exploration of coordinated program changes. The monotone reward-tail map preserves rankings but compresses reward differences, which matters for unnormalized advantages; its causal effect is untested. Additional seeds and controlled comparisons are needed to separate these explanations. Their absence limits the interpretation (Section 5).

F Audit battery and exploit protocol

The adversarial battery checks all coordinate frames and analytic removal volumes. It probes access failures and tool compatibility. Separate tests cover geometric boundaries and reward attribution. Load behavior and entry domains receive dedicated checks. The final group verifies document round trips and malformed-program rejection. The submission receipt records 166 passing checks. The separately implemented 2.5D stock checker matches 98 applicable plan comparisons exactly. It shares motion expansion and path discretization with the voxel checker; these comparisons do not independently validate load, all controller behavior, or multi-face execution. Ten overloaded multi-tool teachers are excluded from that comparison. Four overloaded manual side-face plans are also excluded.

Resolution aliasing. A legal 0.08 mm film passes all grids. A 0.16 mm film is invisible at the 0.2 mm training grid but fails at the routine 0.1 mm audit and the 0.05 mm spot grid. A near-spec construction can appear clean on both fixed grids. Randomized spot selection detects the planted case.

Volumetric and interpolation attacks. Clearance alone can hide tall thin residual slivers, so completion also uses worst residual height. Coarse arc chords can undercut corners. Error-bounded sampling prevents denser sampling from buying reward.

Reward ordering and load padding. The valid objective's affine map reserves a strict margin above malformed output. Executable first-operation prefixes across 20 generated tasks test this ordering. Force-window MRR divides by the union of newly removed voxel-center contact intervals. Zero-removal recuts must leave measured utilization unchanged. Cleared prefixes and equivalent G1 segmentations provide separate checks.

Post-training protocol. During GRPO, a 0.05 mm grade replaces the coarse reward when a geometry-clean candidate is faster than its reference. Final evaluation grades every output at 0.2 and 0.1 mm. Every audit-clean candidate is then rechecked at 0.05 mm. Outputs that already fail at 0.1 mm retain that reward and verdict. Any new discrepancy is minimized and assigned an exploit class. It is then added to the battery without reusing the affected split for selection.

G Compute and reproducibility

The public release is available at `https://github.com/pqori/CNC-RLVR`. The measurement uses one AWS p5.48xlarge node with eight 80 GB H100 GPUs. SFT and GRPO both use one process per GPU; GRPO colocates vLLM with each data-parallel policy replica. The supplement includes the frozen Python package lock and the benchmark hashes. It also includes training code and archived outputs. Model checkpoints and complete wall-clock accounting are not included. Reproducing model generation therefore requires rerunning SFT and GRPO. The reported outputs can be regraded offline.

Physical scope. The checker omits fixture geometry and machine dynamics. No machine trials were conducted. A passing grade does not authorize physical execution.

H Author checklist

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